

Spring 2024 - 2025, Final Examination
 Ordinary Differential Equations (Intl. Session)
 Zhejiang University

Course No.: **MATH1137F**

Exam Date: **2025/04/12 08:00–10:00**

Format: ✓ Closed-Book / Open-Book

Instructions:

- NO notes/calculators or any other form of aid is allowed.
- In order to receive credit, you must **show all of your work**.

Name: _____ **Student ID:** _____

Section	I	II	III	IV	BONUS	Total
Score						
Grader						

I. (10 pts each problem; 30 Total.) :

Score

1. Find the general solution for

$$(2xy^2 - y)dx + (3y^3 + x)dy = 0, \quad x > 0.$$

SOLUTION:

$$\frac{M_y - N_x}{-M} = -\frac{2}{y} \quad \dots\dots\dots 3 \text{ pts}$$

indicates the integrating factor

$$\mu(y) = \frac{1}{y^2}. \quad \dots\dots\dots 2 \text{ pts}$$

Multiplying the original equation to have

$$2xdx + 3ydy + \frac{xdy - ydx}{y^2} = 0 \quad \dots\dots\dots 3 \text{ pts}$$

So the general solution is

$$x^2 + \frac{3}{2}y^2 - \frac{x}{y} = C. \quad \dots\dots\dots 2 \text{ pts}$$

Score

2. Find the general solution to

$$y = 2xy' - 3(y')^2$$

and determine whether it has singular solutions.

SOLUTION:

Let $p = y'$ and differentiate both sides to get

$$pdx = 2pdx + 2xdp - 6pdp \quad \dots\dots\dots 3 \text{ pts}$$

Re-arrange to get

$$\frac{dx}{dp} + \frac{2x}{p} = 6 \quad \dots\dots\dots 2 \text{ pts}$$

Solve to have

$$x(p) = 2p + \frac{C}{p^2}$$

$$y(p) = p^2 + \frac{2C}{p} \quad \dots\dots\dots 3 \text{ pts}$$

$y = 0$ is a singular solution. \dots\dots\dots 2 pts

Score

3. Solve the equation $y'' = 2yy'$ with initial conditions $y(0) = 1$, $y'(0) = 2$.

SOLUTION:

Let $p = \frac{dy}{dx}$ and the original equation becomes

$$p \frac{dp}{dy} = 2py. \quad \dots\dots\dots 3 \text{ pts}$$

So

$$p = \frac{dy}{dx} = y^2 + C \quad \dots\dots\dots 2 \text{ pts}$$

Separate variables, integrate and apply the initial condition to finally get

$$y = C_1 \tan(C_1 x + C_2) \quad \dots\dots\dots 3 \text{ pts}$$

The initial conditions suggest $C_1 = 1$, $C_2 = \frac{\pi}{4}$.

$\dots\dots\dots 2 \text{ pts}$

II. (10 pts each problem; 30 Total.)

Score

1. Find the general solution to $x^2 y'' - 5xy' + 9y = x \ln x$;

SOLUTION:

Characteristic values are $\lambda_1 = \lambda_2 = 3. \quad \dots\dots\dots 3 \text{ pts}$

Setting $y^* = (A \ln x + B)x$

OR use the variation of constants and solve

$$\begin{pmatrix} x^3 & x^3 \ln x \\ 3x^2 & x^2(3 \ln x + 1) \end{pmatrix} \begin{pmatrix} u'_1 \\ u'_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{\ln x}{x} \end{pmatrix} \quad \dots\dots\dots 3 \text{ pts}$$

to have

$$A = B = \frac{1}{4} \quad \text{or} \quad u'_1(t) = -\frac{\ln^2 x}{x^3}, \quad u'_2(t) = \frac{\ln x}{x^3}. \quad \dots\dots\dots 2 \text{ pts}$$

Finally

$$y(x) = (C_1 + C_2 \ln x)x^3 + \frac{x}{4}(1 + \ln x) \quad \dots\dots\dots 2 \text{ pts}$$

Score

2. Find the general solution to $(1 - x^2)y''' + 2xy'' - 2y' = 0$;

SOLUTION:

Let $z = y'$ $z_1(x) = x$ 3 pts

The other fundamental solution can be found via

$$z_2(x) = z_1 \int \frac{e^{-\int \frac{2x}{1-x^2} dx}}{z_1^2(x)} dx = -1 - x^2$$
 3 pts

The general solution is then

$$z(x) = C_1x + C_2(x^2 + 1).$$
 2 pts

$$y(x) = C_1x^2 + C_2(x^3 + 3x) + C_3.$$
 2 pts

Score

3. Find the first three nonzero terms of each of the two linearly independent Frobenius series solutions for the equation

$$2x^2y'' + x(x + 1)y' - 2(x + 1)y = 0.$$

SOLUTION:

Substituting $y = \sum_{n=0}^{\infty} a_n x^{n+r}$ into the original equation we have:

$$2r(r - 1) + r - 2 = 0 \implies r_{1,2} = \frac{1 \pm \sqrt{17}}{4}$$
 3 pts

and

$$\sum_{i=1}^n \{ [2(n+r)(n+r-1) + n+r-2]a_n + (n+r-3)a_{n-1} \} x^{n+r} = 0$$
 3 pts

$$a_n = \frac{(3 - n - r)a_{n-1}}{[2(n+r) + 1](n+r-2)}, \quad n = 1, 2, \dots$$
 3 pts

$$\text{For } r = \frac{1 - \sqrt{17}}{4}, \quad a_1 = -\frac{5 + 3\sqrt{17}}{16}a_0, \quad a_2 = -\frac{9 + 2\sqrt{17}}{26}a_1;$$

$$\text{For } r = \frac{1 + \sqrt{17}}{4}, \quad a_1 = -\frac{5 - 3\sqrt{17}}{16}a_0, \quad a_2 = -\frac{9 - 2\sqrt{17}}{26}a_1$$

..... 1 pts

III. (15 pts each problem; 30 Total.)

Score

1. Consider a system of two species with populations x and y respectively and their dynamics is governed by the following autonomous system:

$$\begin{cases} \frac{dx}{dt} = x^2 - 2x - xy \\ \frac{dy}{dt} = y^2 - 4y + xy \end{cases}$$

(1). Find all the critical points of the system.

SOLUTION:

(0, 0) (2, 0) (0, 4) (3, 1) 4 pts

(2). Characterize each of the critical points you have found in (1) as to type (node/spiral/saddle/center/etc.) and stability (stable/unstable/asymptotically stable).

SOLUTION:

$$\mathbf{J}_{(x,y)} = \begin{pmatrix} 2x - 2 - y & -x \\ y & 2y - 4 + x \end{pmatrix} \quad \dots\dots\dots 3 \text{ pts}$$

(0, 0) : $\lambda = -2, -4$; Improper (2-way) node; Asymptotically stable; 2 pts

(2, 0) : $\lambda = \pm 2$; Saddle; Unstable. 2 pts

(0, 4) : $\lambda = -6, 4$; Saddle; Unstable; 2 pts

(3, 1) : $\lambda = 2 \pm i\sqrt{2}$; Spiral; Unstable; 2 pts

Score

2. Find the solution to the system

$$\begin{cases} dx/dt = 3x - 2y + 15, \\ dy/dt = -x + 3y - 2z, \\ dz/dt = -y + 3z. \end{cases}$$

SOLUTION:

Eigenvalues:

$$\lambda_1 = 1, \lambda_2 = 3, \lambda_3 = 5.$$

..... 3 pts

Associated eigenvectors:

$$\vec{v}_1 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}, \quad \vec{v}_3 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}.$$

..... 6 pts

Then use the variation of parameters to solve

$$\begin{pmatrix} 2e^t & -2e^{3t} & 2e^{5t} \\ 2e^t & 0 & -2e^{5t} \\ e^t & e^{3t} & e^{5t} \end{pmatrix} \vec{u}'(t) = \begin{pmatrix} 15 \\ 0 \\ 0 \end{pmatrix}.$$

or set the particular solution to be $(A, B, C)^T$.

..... 3 pts

$$A = -7, \quad B = -3, \quad C = -1.$$

..... 3 pts

The general solution is therefore:

$$X(t) = \sum_{i=1}^3 C_i X_i(t) = C_1 \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} e^t + C_2 \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} e^{3t} + C_3 \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} e^{5t} - \begin{pmatrix} 7 \\ 3 \\ 1 \end{pmatrix}.$$

Score

IV. (10 Pts) Suppose that the functions $u(x)$ and $v(x)$ are two linearly independent solutions to the second order homogeneous, linear differential equation:

$$y'' + p(x)y' + q(x)y = 0$$

in which $p(x)$ and $q(x)$ are continuous functions defined on the real line.

Prove: $u(x)$ and $v(x)$ don't have common zeroes. Namely, there doesn't exist a $x^* \in \mathbb{R}$ such that $u(x^*) = v(x^*) = 0$. (HINT: Prove first if $u(x^*) = 0$, $u'(x^*) \neq 0$.)

SOLUTION:

Suppose they have a common zero at $x = x^*$

$$W[u, v](x^*) = \begin{vmatrix} u(x^*) & v(x^*) \\ u'(x^*) & v'(x^*) \end{vmatrix} = 0 \quad \dots\dots\dots 5 \text{ pts}$$

But if u and v are linearly independent solutions, their Wronskian should be non-zero everywhere. Thus the contradiction. \dots\dots\dots 5 pts

ALTERNATIVE PROOF:

If $u(x^*) = 0$, $u'(x^*) \neq 0$, since $u \equiv 0$ otherwise; same for v . \dots\dots\dots 5 pts

Consider the non-trivial linear combination

$$\varphi(x) = u'(x^*)v(x) - v'(x^*)u(x)$$

This is also a solution but $\varphi(x^*) = \varphi'(x^*) = 0$. Thus the contradiction. \dots\dots\dots 5 pts

Score

BONUS (5 Pts) Use the Feynman's integration technique to prove

$$\int_0^1 \frac{\ln(1+x)}{1+x^2} dx = \frac{\pi \ln 2}{8}.$$

SOLUTION:

Introduce: $I(\alpha) = \int_0^1 \frac{\ln(1+\alpha x)}{1+x^2} dx, \quad I(0) = 0. \quad \dots\dots\dots 2 \text{ pts}$

Therefore:

$$I'(\alpha) = \int_0^1 \frac{x}{(1+\alpha x)(1+x^2)} dx = \frac{\pi}{4} \frac{\alpha}{1+\alpha^2} + \frac{\ln 2}{2} \frac{1}{1+\alpha^2} - \frac{\ln(1+\alpha)}{1+\alpha^2}. \quad \dots\dots\dots 2 \text{ pts}$$

Integration from 0 to 1 gives:

$$I(1) = \frac{\pi \ln 2}{4} - I(1) \implies I(1) = \frac{\pi \ln 2}{8} \quad \dots\dots\dots 1 \text{ pts}$$